Hugo Armando Guillén-Ramírez

Curriculum Vitae

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Objective

I am PhD student working with machine learning applied to RNA bioinformatics. My interests are mostly algorithm design, machine learning, bioinformatics, optimization using bio-inspired strategies, and graph theory applied to bioinformatics.

Education

2014–Current **PhD in Computer Science**, *CICESE Research Center*, Ensenada, Baja California, Mexico, Grade: 97.02/100.

2011–2014 **MSc in Computer Science**, *CICESE Research Center*, Ensenada, Baja California, Mexico, Grade: 96.78/100.

2007–2011 **Bachelor in Computer Systems Engineering**, *Instituto Tecnológico de Tepic*, Nayarit, Mexico, Grade: 97/100.

Doctoral thesis (ongoing)

Title *Methods for non-coding RNA gene prediction* (translated from spanish)

Description The main objective of this project is to develop a non-coding RNA (ncRNA) gene predictor based in homology search on known sequences performed by classifiers, meta-heuristics and state-of-the-art biological information on this kind of molecules.

Tentative end date January 2018

Master thesis

Title *Design of a Tissue P-System and a molecular algorithm to solve the MAX-CLIQUE problem* (translated from spanish)

Description This theoretical work presents a tissue P-System that solves the MAX-CLIQUE problem in polynomial time, and also a constructive DNA algorithm inspired in this system.

Publications

Conference **Hugo A. Guillen-Ramirez**, Jose Colbes, Carlos A. Brizuela, Israel M. Martinez-Perez. "Accurate Classification of Immunomodulatory RNA Sequences". In International Joint Conference on Neural Networks (IJCNN) 2017. pp. 236-241. IEEE, 2017. [Accepted, publication pending.]

Poster **Hugo A. Guillen** and Israel Martinez-Perez. "Simulation of a programmable molecular computer". Presented in the 21st International Conference on DNA Computing and Molecular Programming 2015, Boston, USA.

Conference **Hugo A. Guillen** and Israel Martinez-Perez. "Challenges in the prediction of non-coding RNA genes" (translated from spanish). In the proceedings of the National Meeting on Computer Science (Encuentro Nacional de Ciencias de la Computación, ENC 2014), Mexico.

Conference Brizuela, Carlos A., Jorge E. Luna-Taylor, Israel Martinez-Perez, **Hugo A. Guillen**, David O. Rodriguez, Armando Beltran-Verdugo. "Improving an evolutionary multi-objective algorithm for the biclustering of gene expression data". In Evolutionary Computation (CEC), 2013 IEEE Congress on, pp. 221-228. IEEE, 2013.

Poster **Hugo A. Guillen** and Israel Martinez-Perez. "Design of a Tissue P-System for solving the MAX-CLIQUE problem" (translated from spanish). Presented in the 6th International Congress on Computer Science (6to.Congreso Internacional en Ciencias Computacionales, CiComp 2013), Mexico.

Experience

Talks

2017, Mar Seminar **Score functions for discriminant k-mers and their application in ncRNA classification** (translated from spanish) at CICESE Research Center, Mexico.

2016, Oct Seminar **Methods for ncRNA gene prediction: RNA secondary structure alternative graph representations** (translated from spanish) in the XII National Meeting on Mathematic Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2016, Oct Hands-on **Bio-computing: RNA, BioPython, and Graphs** (translated from spanish) in the Computer Science Fall School at CICESE Research Center, Mexico.

2016, Aug Primer course on **Discrete Mathematics** (translated from spanish) in the Computer Science Department at CICESE Research Center, Mexico.

2016, Jun Seminar **Developing scientific discoverability online** (translated from spanish) at CICESE Research Center, Mexico.

2015, Dec Seminar **Selected topics on biocomputing** (translated from spanish) at Instituto Tecnológico de Tepic, Mexico.

2015, Aug Primer course on **Discrete Mathematics** (translated from spanish) in the Computer Science Department at CICESE Research Center, Mexico.

2015, Mar Seminar **Challenges in ncRNA gene prediction** (translated from spanish) at CICESE Research Center, Mexico.

2014, Nov Seminar **Challenges in ncRNA gene prediction** (translated from spanish) in the National Meeting on Computer Science (Encuentro Nacional de Ciencias de la Computación 2014), Mexico.

2012, May Seminar **Introduction to genetic algorithms** (translated from spanish) at Instituto Tecnológico de Tepic, Mexico.

Attendant

2016 XVIII Fall School on Mathematic Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2016 NGS Data Analysis and Informatics Conference, San Diego, USA.

2015 Applications of Artificial Organic Networks (National Meeting on Computer Science 2015, Mexico).

Teaching

2012 Teacher of the **Computer Architecture** course at Professional Development University(UNIDEP) campus Ensenada, Mexico.

2012 T.A. of the **Discrete Mathematics** course at CICESE Research Center.

Developer

2012 MXGlobal Solutions - Tepic, Nayarit, Mexico. I developed C# and SQL software.

2010-211 Instituto Tecnológico de Tepic - Tepic, Nayarit, Mexico. I developed a web system in Java/JSP.

Scholarships

2014 CONACYT National Scholarship for doctoral studies (Mexico).

2011 CONACYT National Scholarship for master studies (Mexico).

Skills

Advanced PYTHON, JAVA, C#, Weka 3

Intermediate SQL (PostgreSQL, MySQL, SQLite), JAVA, PHP, XHTML, CSS, JAVASCRIPT, LaTeX, LINUX

Basic C++, VISUALBASIC .NET

Languages

Spanish Native

English Advanced

Other activities

2016 Developer of REFRACTIVEINDEX.INFO-SQLITE, a Python 3 open source package to retrieve information from the refractiveindex.info database locally. `http://github.com/HugoGuillen/refractiveindex.info-sqlite`

2015 Adjunct supervisor of Raquel Garza Gómez in the project "Development of auxiliary Python scripts to process non-coding RNA databases" (translated from spanish) on August-December 2015, at CICESE Research Center, Mexico.

2004–2006 Participant of the XVIII, XIX, and XX Mexican Mathematical Olympiads.

References

Available at request.